Serial star-gobbling black hole
Astronomers have detected a supermassive black hole that consumes a sun-sized star every two days. **https://dgit.in/Jun18-B34-1**

World's deepest plastic bag
A plastic bag was found in the deepest point in the ocean, the Mariana Trench, no place is safe from pollution. **https://dgit.in/Jun18-B34-2**

Hawking's final hurrah!

We try and understand what Prof. Stephen Hawking's last paper was about, and how significant it might (or might not) be for the next decade in science

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It's very rare for a given year to be remembered by someone's birth. Instead, it's usually death, tragedy, or mind-blowing accomplishments that rule the zeitgeist of the year. Unless it's a royal family's child, or the first test tube baby – Louise Brown, born on July 25, 1978 – it's very, very rare for people to be born famous. Thus, it's a rather macabre pastime in the age of the internet to remember calendar years by who died, and we're going to shamelessly partake in it. For us, no matter what happens from here on out, 2018 will always be the year the world lost one of its greatest physicists.

Professor Stephen Hawking may have had a body that was broken, but he had a mind that was anything but. His dedication to science (physics) earned him accolades, his papers made him famous, and his ability to overcome his horrific disability was inspirational to the world, but it was his sense of humour and eagerness to participate in culture that made him a household name across most of the connected world.

Prof. Stephen Hawking – his mind literally knew no boundaries!

While we ourselves have written enough and more about him in the past, especially the cultural aspect, in this space we want to focus on his work, more specifically his final paper in physics.

A SMOOTH EXIT FROM ETERNAL INFLATION?

Published a little over a month after Hawking passed away, his final 17 page paper was written in collaboration with famous Belgian physicist Thomas Hertog, and turned out to be the final time Hawking would look back at his famous No Boundary theory. The mark of a great scientist is one who isn't burdened by confirmation bias, or takes too much pride in their own theories. His final paper is actually a departure from his beloved No Boundary hypothesis – Hertog was recently quoted as saying, "The paper represents indeed a departure from the no-boundary theory (as we write at the end)."

But first, let's take a quick (and a very layman-level) dive into what the No Boundary theory is all about.

If you've been following our science coverage, or reading about the Big Bang, you know that there's a problem when you extrapolate backwards to the event itself. Physicists are very confident that the Big Bang happened, and feel they're capable of extrapolating knowledge back to moments after the Big Bang, but the event itself is something we just cannot apply the laws of physics to. This is because after Einstein, time became an integral part of all physics, and as we know, we measure rate of change of anything in our universe as rate of change over time. How can we ever reach a point where time is zero? Since anything divided by zero is undefined (not infinity, as many mistakenly still believe), we just cannot understand what it means to have a time = zero.

Imagine if time went in reverse, and you were somehow outside the bounds of the universe, looking in.